Roll No. ....

Z–8–C

Total No. of Questions : 4] [Total No. of Printed Pages : 8

11th ARF(SZ)JKUT2024–25

208–C

PHYSICS

Time : 3 Hours] **[Maximum Marks : 70**

Section–A

(Objective Type Questions) 1 each

1. (i) The physical quantities not having same dimensions are :

(A) Momentum and Planck's constant

(B) Speed and $(\mu_0 \epsilon_0)^{-1/2}$

(C) Speed and $\sqrt{P/\rho}$

(D) Surface tension and spring constant

(ii) The number of significant figures in 0.2370 is :

(A) 1 (B) 3

(C) 4 (D) 5

(iii) Which among the following can be zero when a particle is in motion for some time ?

(A) Displacement

(B) Speed

(C) Distance

(D) None of these

(iv) The path traced by a projectile in space is known as :

(A) Track

(B) Orbit

(C) Trajectory

(D) Coral

(v) Which one of the following is not an example of the third law of motion ?

(A) Walking

(B) Cycling

(C) Skiing

(D) Walking on a boat

(vi) Which of the following is self adjusting force ?

(A) Kinetic friction

(B) Limiting friction

(C) Static friction

(D) All of these

(vii) A body in rotational motion possesses rotational kinetic energy given by :

(A) $\frac{1}{2}I\omega^2$

(B) $\frac{1}{2}I^2\omega$

(C) $2\,I^2\omega$

(D) $I\omega$

(viii) Which law describes the orbits of planets around the sun ?

(A) Newton's law

(B) Kirchhoff's law

(C) Faraday's law

(D) Kepler's law

(ix) Wien's law is :

(A) $\lambda_m T^{-1}$ = constant

(B) $\lambda_m T$ = constant

(C) $\frac{\lambda_m}{T}$ is constant

(D) $\frac{\lambda_m}{\sqrt{T}}$ = constant

(x) How many number of particles are in 22.4 litres of any gas at S.T.P. ?

(A) 6.02×10^{22}

(B) 2.06×10^{23}

(C) 6.02×10^{23}

(D) 2.06×10^{22}

Section–B

(Very Short Answer Type Questions) 2 each

2. (a) From the velocity time graph of uniform accelerated motion, deduce the equation of motion in distance and velocity.

(b) Explain the term impulse. Give *one* example.

(c) A force $F = (10 + 0.50x)$ acts on a particle in x direction, where F is in newton and x is in metre. Find the work done by this force during a displacement from $x = 0$ to $x = 2$ m.

(d) How do potential energy of a spring vary with displacement ?

(e) What is Collision ? Define *two* types of collision.

(f) A torque of 20 N-m is applied on a wheel initially at rest. Calculate angular momentum of the wheel after 3 seconds.

(g) State universal law of gravitation. Derive relation between 'g' and 'G'.

(h) State law of equipartition of energy. Also give the concept of mean free path.

(i) Why longitudinal waves are called pressure waves ?

Section–C

(Short Answer Type Questions) 3 each

3. (a) The amplitude of a particle executing SHM is 0.1 m. The acceleration of the particle at a distance of 0.03 m from equilibrium position is 0.12 m/s^2. What will be its velocity at a distance of 0.08 m from position of equilibrium ?

(b) Find the dimensions of Planck's constant. If its value in CGS system is 6.62×10^{-27} erg-sec. What will be its value on MKS system ?

(c) Why are the curved roads banked ? Find the angle of banking of curved road.

(d) State and explain the principle of conservation of angular momentum.

(e) Define orbital velocity. Derive an expression for it.

(f) State and explain Hooke's law.

(g) Define Thermal Equilibrium. How does it occur ? What is the relationship between thermal equilibrium and zeroth law ?

(h) State second law of thermodynamics by Kelvin and Clausius. Give *two* applications of second law of Thermodynamics.

(i) Show that average kinetic energy of translation per molecule of gas is directly proportional to the absolute temperature of gas.

Section–D

(Long Answer Type Questions) 5 each

(a) Show that the path followed by projectile is parabolic when it is projected at an angle θ with horizontal.

Or

Explain cross product of vectors. Give its example and properties.

(b) State and prove Bernoulli's theorem. Give its applications.

Or

Define Capillarity. Derive ascent formula for the rise of a liquid in a capillary tube and discuss the result obtained.

(c) What are Standing Waves ? Explain their formation in open organ pipe.

Or

Show that the total energy of particles performing linear simple harmonic motion is constant. What is the time period of total energy in SHM ?